

a Four preregistered Replogle generalization regimes

Seen response context

IID
K562 · training targets

perturbation-condition bootstrap

Perturbation OOD
K562 · sealed targets

perturbation-condition bootstrap

Unseen response context

Context OOD
RPE1 · known targets

perturbation-condition bootstrap

Double OOD
RPE1 · sealed targets

perturbation-condition bootstrap

Seen target

Unseen target

b

One-shot Adamson external confirmation

protocol
committed

27 target
predictions frozen

terminal
decision



controls only
5,241 cells

outcomes opened
exactly once

55 disjoint reference targets estimate systematic perturbed variation · no post-outcome tuning